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From quality assurance to quality engineering: a multi-agent, local-first approach for agentic DevOps

Supervisors

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Background and motivation

Software quality has traditionally been addressed through Quality Assurance (QA) practices applied at discrete checkpoints in the development lifecycle. The emergence of DevOps and continuous delivery pipelines introduced the idea that quality must be a continuous, first-class concern embedded throughout the entire development cycle — a paradigm shift captured by the concept of Quality Engineering (QE).

Generative AI and autonomous agents are now transforming the software engineering toolchain. Large Language Models (LLMs) and agent frameworks (e.g., LangChain, AutoGen) are being applied not only to code generation and review but also to test generation, static analysis, vulnerability detection, and CI/CD pipeline management. These agentic approaches introduce a new class of automation with “actors” capable of reasoning, tool use, and multi-step decision-making.

Cloud-based agentic solutions present significant drawbacks for widespread industrial adoption: high operational costs driven by per-token LLM usage, data-privacy concerns when proprietary code is sent to external APIs, and latency incompatible with fast feedback loops. Recent advances in Small Language Models (SLMs) and hybrid local/cloud orchestration strategies offer a promising path to address these constraints, enabling cost-efficient, privacy-preserving agentic quality engineering.

There is, however, a gap in systematic approaches that integrate agentic QE practices into real DevOps pipelines, particularly those leveraging local-first or hybrid SLM/LLM architectures and validated against real industrial systems. This dissertation aims to address that gap.

Objectives

This dissertation has three main objectives:

- Design and implement a multi-agent architecture for Quality Engineering, covering the orchestration of specialised agents that perform tasks such as test generation, static analysis, and vulnerability detection, treating quality as a first-level concern throughout the development lifecycle.
- Develop local-first and hybrid SLM/LLM strategies that reduce operational costs and address data-privacy constraints, while maintaining effectiveness comparable to fully cloud-based approaches.
- Integrate and validate the proposed solution within real DevOps pipelines, using case studies from existing systems in collaboration with UA’s IT department (STIC), and evaluate the solution’s impact on software quality metrics and development cycle.

Work plan

The dissertation is structured across five phases as depicted in the following high-level plan.

Phase	Activities	Outcomes
1 — Literature Review & Problem Framing	Review of agentic QE, SLM/LLM benchmarks, and DevOps quality tooling; definition of evaluation criteria and research questions	Literature review report; refined research questions
2 — Architecture Design	Design of the multi-agent QE architecture; selection of agent frameworks; design of local-first/hybrid orchestration strategies; identification of STIC use cases	Architecture design document; use case catalogue
3 — Prototype Development	Implementation of the agent orchestration layer; integration of SLM/LLM backends (local and cloud); development of QE agents (test generation, static analysis, vulnerability detection)	Working prototype; component integration tests
4 — Integration & Evaluation	Deployment into STIC DevOps pipelines; execution of case studies; collection of quality metrics (defect detection rate, cost, latency, coverage); comparison with baseline approaches	Evaluation report; validated case study results
5 — Dissertation Writing	Writing and revision of the dissertation document; tentative: preparation of one scientific paper for submission	Dissertation.

References

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